DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

WO-300-01 — Pull, clean and test the tube bundle of E-101

Area	A300 Exchangers and drum	**Task**	XB Exchanger bundle pull, clean and tube-side test
Priority	2 (integrity)	**Issued by**	Facility owner (demonstration)
Primary asset	E-101 — E-101 shell-and-tube exchanger 36" ID × 192" — E 111'-6" N -25'-0" EL 106'-6"		
Equipment	E-101	**Lines**	4"-P300-P-1003, 4"-P300-P-1005, 4"-P150-CWS-3001, 4"-P150-CWR-3002, 1"-P150-V-6021, 1"-P150-D-6022, 3/4"-P300-I-6023
Isometrics	ISO-P-1003, ISO-P-1005, ISO-CWS-3001, ISO-CWR-3002, ISO-V-6021, ISO-D-6022, ISO-I-6023 (owner's drawings, attached)		
Permits	General work permit, Confined-space entry permit, Lifting plan		

Reason

Fouling: tube bundle to be pulled, cleaned and tested.

Scope of work

1. Isolate E-101 per the isolation list (tube and shell sides); drain and vent.
2. Remove the channel cover; pull the tube bundle to the cleaning bay.
3. Clean the tubes and shell side; inspect visually; thickness at the TMLs.
4. Reinstall the bundle with new gaskets; tube-side hydrostatic test at 420 psi (SW-TEST-1).
5. Remove blinds; restore the line-up.

Affected items: E-101

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-TEST-1, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-6, SW-WO-7

Owner's isometric drawings (SW-IMP-6)

Drawing	Line drawn	Work line	Sheets	Match to the plant model
ISO-P-1003	4"-P300-P-1003	4"-P300-P-1003	3	3 error(s), 3 warning(s) — see below
ISO-P-1005	4"-P300-P-1005	4"-P300-P-1005	4	0 error(s), 5 warning(s) — see below
ISO-CWS-3001	4"-P150-CWS-3001	4"-P150-CWS-3001	1	agrees with the model
ISO-CWR-3002	4"-P150-CWR-3002	4"-P150-CWR-3002	1	agrees with the model
ISO-V-6021	1"-P150-V-6021	1"-P150-V-6021	1	agrees with the model
ISO-D-6022	1"-P150-D-6022	1"-P150-D-6022	1	agrees with the model
ISO-I-6023	3/4"-P300-I-6023	3/4"-P300-I-6023	1	agrees with the model

ISO-P-1003 (PDF): ../isometrics-imported/ISO-P-1003.pdf · ISO-P-1005 (PDF): ../isometrics-imported/ISO-P-1005.pdf · ISO-CWS-3001 (PDF): ../isometrics-imported/ISO-CWS-3001.pdf · ISO-CWR-3002 (PDF): ../isometrics-imported/ISO-CWR-3002.pdf · ISO-V-6021 (PDF): ../isometrics-imported/ISO-V-6021.pdf · ISO-D-6022 (PDF): ../isometrics-imported/ISO-D-6022.pdf · ISO-I-6023 (PDF): ../isometrics-imported/ISO-I-6023.pdf — the sheets that hold the work follow the ITP.

Drawing	Sheet	Level	Finding
ISO-P-1003	1	error	sheet 1: CONT'D ON ISO-P-1004 prints E 154'-3 5/16" N -17'-6" EL 107'-6" for E 154'-3 5/16" N -16'-6" EL 107'-6": the minus sign applies to the feet only; read as written the point is 12.00 in away and 12.00 in off where 4"-P300-P-1003 meets 4"-P300-P-1004 (E 154'-3 5/16" N -16'-6" EL 107'-6")
ISO-P-1003	2	error	sheet 2: CONT'D ON ISO-P-1003 SH 3 prints E 110'-6" N -4'-9" EL 118'-6" for E 110'-6" N -3'-3" EL 118'-6": the minus sign applies to the feet only; read as written the point is 18.00 in away (still on the centreline, at the wrong place along it)
ISO-P-1003	3	error	sheet 3: CONT'D FROM ISO-P-1003 SH 2 prints E 110'-6" N -4'-9" EL 118'-6" for E 110'-6" N -3'-3" EL 118'-6": the minus sign applies to the feet only; read as written the point is 18.00 in away (still on the centreline, at the wrong place along it)
ISO-P-1003	3	warning	W18 (4"-P300-P-1003/W18, sheet 3) is labelled again on sheet 2; it is 5.7 in from the nearest break
ISO-P-1003	2	warning	branch weld 3/4"-P300-I-7003/W1 (E 110'-6" N -2'-6" EL 118'-9 5/16", sheet 2) has no label
ISO-P-1003	3	warning	branch weld 3/4"-P300-SP-7022/W1 (E 110'-6" N -4'-0" EL 118'-9 5/16", sheet 3) has no label
ISO-P-1005	1	warning	weld 4"-P300-P-1005/W11 (E 109'-11/16" N -4'-0" EL 116'-6", sheet 1) has no label
ISO-P-1005	3	warning	W18 (4"-P300-P-1005/W18, sheet 3) is labelled again on sheet 2; it is 223.3 in from the nearest break
ISO-P-1005	3	warning	weld 4"-P300-P-1005/W19 (E 34'-11/16" N -2'-0" EL 116'-6", sheet 3) has no label
ISO-P-1005	4	warning	weld 4"-P300-P-1005/W21 (E 33'-6" N -2'-0" EL 117'-11 5/16", sheet 4) has no label
ISO-P-1005	2	warning	branch weld 3/4"-P300-I-7005/W1 (E 109'-0" N -2'-0" EL 116'-9 5/16", sheet 2) has no label

Assets in the plant model (SW-WO-7)

Tag	Kind	Description	Line	Location
E-101	equipment	E-101 shell-and-tube exchanger 36" ID × 192"		E 111'-6" N -25'-0" EL 106'-6"
4"-P300-P-1003/B2	item	VALVE, GATE, WEDGE, OS&Y 4" P300	4"-P300-P-1003	E 110'-6" N -25'-0" EL 109'-3-13/16"

Tag	Kind	Description	Line	Location
E-101-N1	nozzle	Nozzle E-101-N1 (tube in, 4" P300) on E-101		E 110'-6" N -25'-0" EL 108'-8-1/16"
4"-P300-P-1005/A2	item	VALVE, GATE, WEDGE, OS&Y 4" P300	4"-P300-P-1005	E 110'-6" N -25'-0" EL 103'-8-3/16"
E-101-N2	nozzle	Nozzle E-101-N2 (tube out, 4" P300) on E-101		E 110'-6" N -25'-0" EL 104'-3-15/16"
4"-P150-CWS-3001/B2	item	VALVE, GATE, WEDGE, OS&Y 4" P150	4"-P150-CWS-3001	E 113'-6" N -25'-0" EL 109'-3-13/16"
E-101-N3	nozzle	Nozzle E-101-N3 (shell in, 4" P150) on E-101		E 113'-6" N -25'-0" EL 108'-8-1/16"
4"-P150-CWR-3002/A2	item	VALVE, GATE, WEDGE, OS&Y 4" P150	4"-P150-CWR-3002	E 125'-6" N -25'-0" EL 103'-8-3/16"
E-101-N4	nozzle	Nozzle E-101-N4 (shell out, 4" P150) on E-101		E 125'-6" N -25'-0" EL 104'-3-15/16"
1"-P150-V-6021/2	item	VALVE, BALL, FLOATING 1" P150	1"-P150-V-6021	E 124'-6" N -25'-0" EL 109'-1/2"
E-101-N5	nozzle	Nozzle E-101-N5 (vent, 1" P150) on E-101		E 124'-6" N -25'-0" EL 108'-5-11/16"
1"-P150-D-6022/2	item	VALVE, BALL, FLOATING 1" P150	1"-P150-D-6022	E 113'-6" N -25'-0" EL 103'-11-1/2"
E-101-N6	nozzle	Nozzle E-101-N6 (drain, 1" P150) on E-101		E 113'-6" N -25'-0" EL 104'-6-3/8"
3/4"-P300-I-6023/2	item	VALVE, BALL, FLOATING 3/4" P300	3/4"-P300-I-6023	E 119'-6" N -27'-6-1/4" EL 106'-6"
E-101-N7	nozzle	Nozzle E-101-N7 (instrument, 0.75" P300) on E-101		E 119'-6" N -26'-11-1/2" EL 106'-6"

Isolation list (SW-WO-3)

#	Kind	Tag	Line	Action	Location	Fitted ✓	Removed ✓
1	valve	4"-P300-P-1003/B2	4"-P300-P-1003	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1003/B2) on E-101-N1	E 110'-6" N -25'-0" EL 109'-3-13/16"		
2	blind	E-101-N1	4"-P300-P-1003	Fit a blind between 4"-P300-P-1003/B2 and the nozzle flange E-101-N1	E 110'-6" N -25'-0" EL 108'-8-1/16"		
3	valve	4"-P300-P-1005/A2	4"-P300-P-1005	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P300 (4"-P300-P-1005/A2) on E-101-N2	E 110'-6" N -25'-0" EL 103'-8-3/16"		
4	blind	E-101-N2	4"-P300-P-1005	Fit a blind between 4"-P300-P-1005/A2 and the nozzle flange E-101-N2	E 110'-6" N -25'-0" EL 104'-3-15/16"		
5	valve	4"-P150-CWS-3001/B2	4"-P150-CWS-3001	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P150 (4"-P150-CWS-3001/B2) on E-101-N3	E 113'-6" N -25'-0" EL 109'-3-13/16"		
6	blind	E-101-N3	4"-P150-CWS-3001	Fit a blind between 4"-P150-CWS-3001/B2 and the nozzle flange E-101-N3	E 113'-6" N -25'-0" EL 108'-8-1/16"		
7	valve	4"-P150-CWR-3002/A2	4"-P150-CWR-3002	Close and lock VALVE, GATE, WEDGE, OS&Y 4" P150 (4"-P150-CWR-3002/A2) on E-101-N4	E 125'-6" N -25'-0" EL 103'-8-3/16"		
8	blind	E-101-N4	4"-P150-CWR-3002	Fit a blind between 4"-P150-CWR-3002/A2 and the nozzle flange E-101-N4	E 125'-6" N -25'-0" EL 104'-3-15/16"		
9	valve	1"-P150-V-6021/2	1"-P150-V-6021	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-V-6021/2) on E-101-N5	E 124'-6" N -25'-0" EL 109'-1/2"		
10	blind	E-101-N5	1"-P150-V-6021	Fit a blind between 1"-P150-V-6021/2 and the nozzle flange E-101-N5	E 124'-6" N -25'-0" EL 108'-5-11/16"		
11	valve	1"-P150-D-6022/2	1"-P150-D-6022	Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-D-6022/2) on E-101-N6	E 113'-6" N -25'-0" EL 103'-11-1/2"		
12	blind	E-101-N6	1"-P150-D-6022	Fit a blind between 1"-P150-D-6022/2 and the nozzle flange E-101-N6	E 113'-6" N -25'-0" EL 104'-6-3/8"		
13	valve	3/4"-P300-I-6023/2	3/4"-P300-I-6023	Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-I-6023/2) on E-101-N7	E 119'-6" N -27'-6-1/4" EL 106'-6"		
14	blind	E-101-N7	3/4"-P300-I-6023	Fit a blind between 3/4"-P300-I-6023/2 and the nozzle flange E-101-N7	E 119'-6" N -26'-11-1/2" EL 106'-6"		

Inspections

Id	Inspection	Criteria	Basis
WO-300-01-I1	Internal visual examination of the shell, channel and tube bundle	Findings recorded against the owner's integrity criteria	SW-NDE-4
WO-300-01-I2	Ultrasonic thickness at the 12 TMLs listed	Each reading above the owner's minimum wall	SW-WO-6

Thickness monitoring locations (SW-WO-6)

TML	Item	Location	Position	Nominal wall	R1	R2	R3	R4
WO-300-01-TML01	E-101	Nozzle neck E-101-N1	E 110'-6" N -25'-0" EL 108'-8-1/16"		top: ____	side: ____	bottom: ____	side: ____
WO-300-01-TML02	E-101	Nozzle neck E-101-N2	E 110'-6" N -25'-0" EL 104'-3-15/16"		top: ____	side: ____	bottom: ____	side: ____
WO-300-01-TML03	E-101	Nozzle neck E-101-N3	E 113'-6" N -25'-0" EL 108'-8-1/16"		top: ____	side: ____	bottom: ____	side: ____

TML	Item	Location	Position	Nominal wall	R1	R2	R3	R4
WO-300-01-TML04	E-101	Nozzle neck E-101-N4	E 125'-6" N -25'-0" EL 104'-3-15/16"		top: ____	side: ____	bottom: ____	side: ____
WO-300-01-TML05	E-101	Nozzle neck E-101-N5	E 124'-6" N -25'-0" EL 108'-5-11/16"		top: ____	side: ____	bottom: ____	side: ____
WO-300-01-TML06	E-101	Nozzle neck E-101-N6	E 113'-6" N -25'-0" EL 104'-6-3/8"		top: ____	side: ____	bottom: ____	side: ____
WO-300-01-TML07	E-101	Nozzle neck E-101-N7	E 119'-6" N -26'-11-1/2" EL 106'-6"		top: ____	side: ____	bottom: ____	side: ____
WO-300-01-TML08	E-101	Left head centre	E 110'-8-3/4" N -25'-0" EL 106'-6"		centre: ____			
WO-300-01-TML09	E-101	Right head centre	E 128'-3-1/4" N -25'-0" EL 106'-6"		centre: ____			
WO-300-01-TML10	E-101	Shell at 0.000" from the left tangent	E 111'-6" N -25'-0" EL 106'-6"		0°: ____	90°: ____	180°: ____	270°: ____
WO-300-01-TML11	E-101	Shell at 6'-0" from the left tangent	E 117'-6" N -25'-0" EL 106'-6"		0°: ____	90°: ____	180°: ____	270°: ____
WO-300-01-TML12	E-101	Shell at 12'-0" from the left tangent	E 123'-6" N -25'-0" EL 106'-6"		0°: ____	90°: ____	180°: ____	270°: ____

Test — Hydrostatic test (tube side)

Description	Tube side of the reinstalled bundle
Pressure	420 psi
Hold	30 minutes, no leak, no pressure drop
Basis	SW-TEST-1

Inspection and test plan (SW-ITP-1..3)

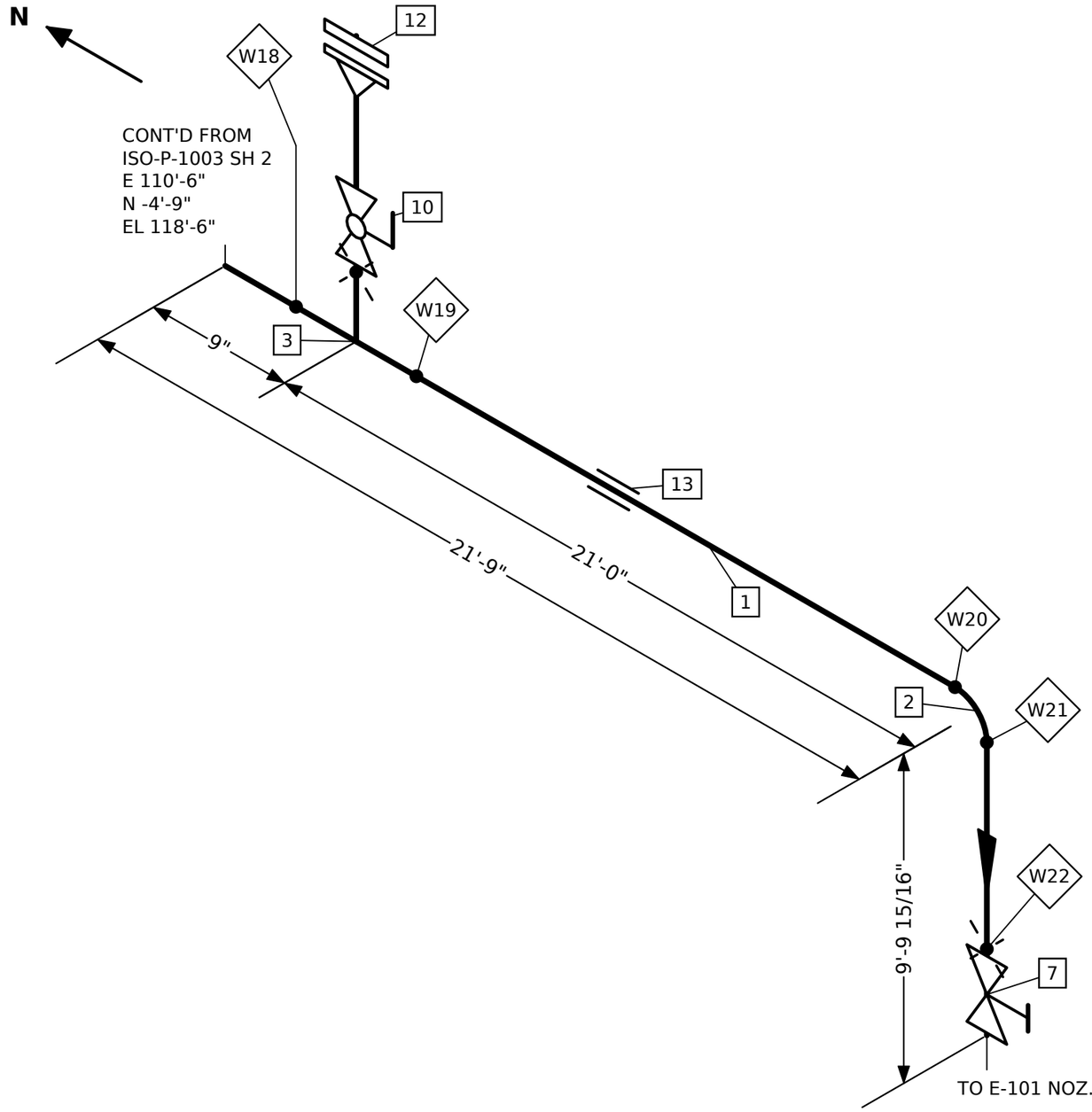
H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

#	Activity	Reference	Acceptance	Record	Contr.	Owner	3rd party	Status	Signed (contr. / owner / 3rd)	Date	Remark
1	Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained	SW-WO-3, isolation list	Every isolation in place and tagged; zero pressure at the vent	Signed isolation certificate	H	H	—				
2	Equipment vented, purged and gas-tested for entry; confined-space entry permit issued	Owner entry procedure	Atmosphere safe for entry per owner limits	Gas test record, entry permit	H	H	—				
3	Channel cover removed; bundle pulled and cleaned	Owner procedure	Surfaces clean enough to inspect	Daily report	S	S	—				
4	Internal visual examination	SW-NDE-4	Findings recorded	Inspection report with photographs	W	H	W				
5	Ultrasonic thickness at 12 TMLs	SW-WO-6	Above the owner's minimum wall	Thickness record per TML	S	R	R				
6	Owner's decision on findings (repair, monitor or accept)	Owner integrity procedure	Decision recorded	Signed disposition	R	H	R				
7	Closed with new gaskets on every opened joint; studs to preload	SW-WO-5	Preload as stated	Bolt-up record	W	W	—				
8	Tube-side hydrostatic test at 420 psi, 30 minutes	SW-TEST-1	No leak, no pressure drop	Test record	H	H	H				
9	Blinds removed, isolations returned to normal line-up per the isolation list	SW-WO-3	Every blind removed and accounted for	Isolation certificate closed	H	H	—				
10	Records complete; work order handed back to operations	SW-ITP-3	All ITP activities accepted or not applicable	Signed ITP and completion record	H	H	R				

Where the work is on the owner's drawings (SW-IMP-8)

Each affected item, replaced weld and isolation point, with the drawing and sheet that hold it and the weld label printed there. The sheets follow, copied unchanged.

Tag	Role	Drawing	Sheet	Weld label
4"-P300-P-1003/B2	isolation: valve	ISO-P-1003	3	
4"-P300-P-1005/A2	isolation: valve	ISO-P-1005	1	
4"-P150-CWS-3001/B2	isolation: valve	ISO-CWS-3001	1	
4"-P150-CWR-3002/A2	isolation: valve	ISO-CWR-3002	1	
1"-P150-V-6021/2	isolation: valve	ISO-V-6021	1	
1"-P150-D-6022/2	isolation: valve	ISO-D-6022	1	
3/4"-P300-I-6023/2	isolation: valve	ISO-I-6023	1	



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	28'-2"	4"	PIPE 4" P300 BEVELLED ENDS
2	1	4"	ELBOW 90° R = 5/3 OD 4" P300
3	1	4"x3/4"	TEE REDUCING 4" × 3/4" P300
4	2	4"	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300
5	1	4"	FLANGE WELD-NECK 4" P300 RAISED FACE
6	2	4"	GASKET 4" P300 COMPRESSED SHEET
7	1	4"	VALVE, GATE, WEDGE, OS&Y 4" P300
8	1	3/4"	FLANGE WELD-NECK 3/4" P300 RAISED FACE
9	2	3/4"	GASKET 3/4" P300 COMPRESSED SHEET
10	1	3/4"	VALVE, BALL, FLOATING 3/4" P300
11	2	3/4"	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300
12	1	3/4"	FLANGE BLIND 3/4" P300 RAISED FACE
13	1	4"	HANGER FOR 4" LINE (HANGER-4x166.0)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

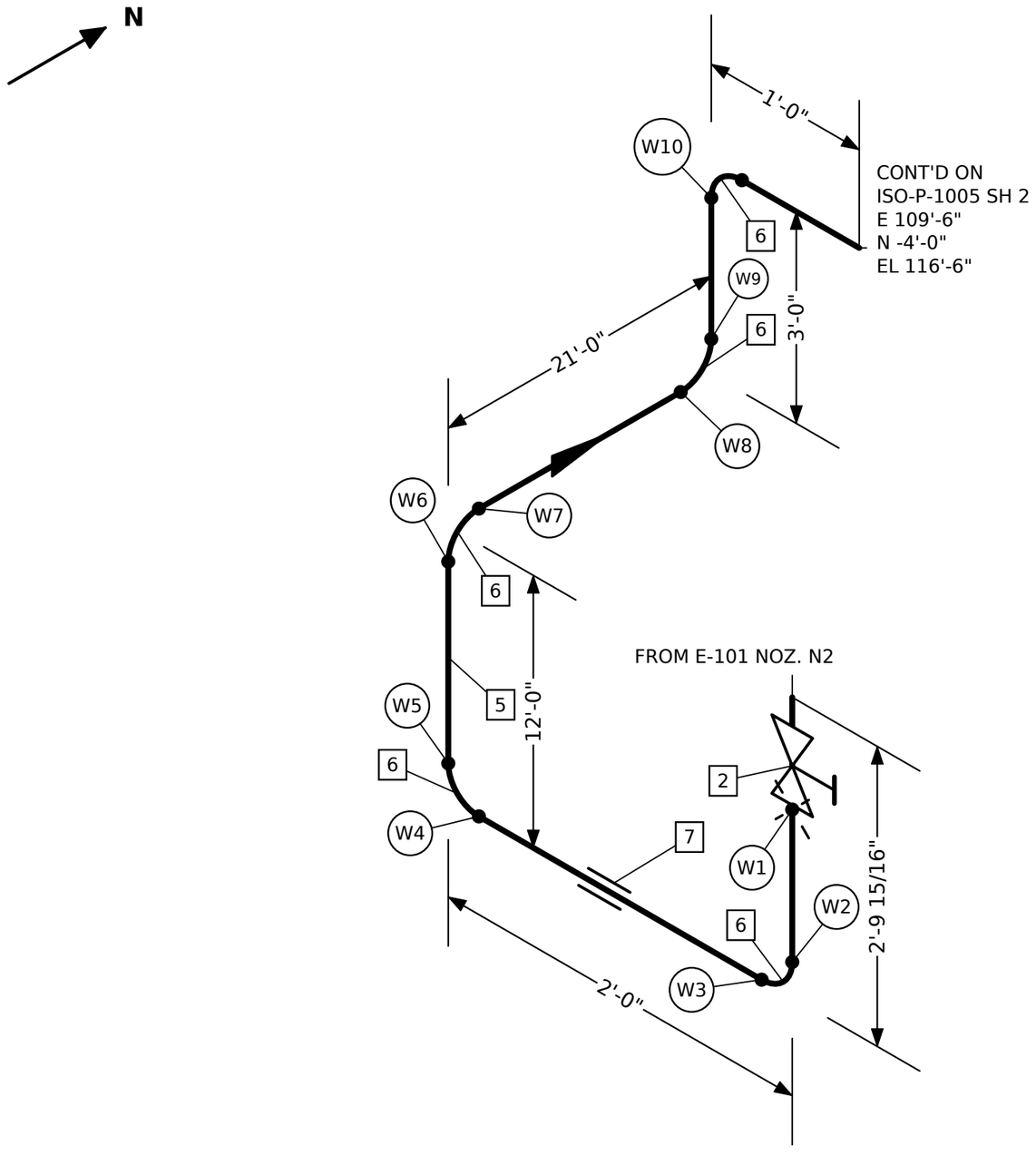
NOTES:
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	P

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
4"-P300-P-1003	H300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-P-1003	3 OF 3	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	2	4"	GASKET 4" P300 COMPRESSED SHEET
2	1	4"	VALVE, GATE, WEDGE, OS&Y 4" P300
3	2	4"	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P300
4	1	4"	FLANGE WELD-NECK 4" P300 RAISED FACE
5	34'-6 5/16"	4"	PIPE 4" P300 BEVELLED ENDS
6	5	4"	ELBOW 90° R = 5/3 OD 4" P300
7	1	4"	TRUNNION FOR 4" LINE (TRUNNION-4x16.0)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

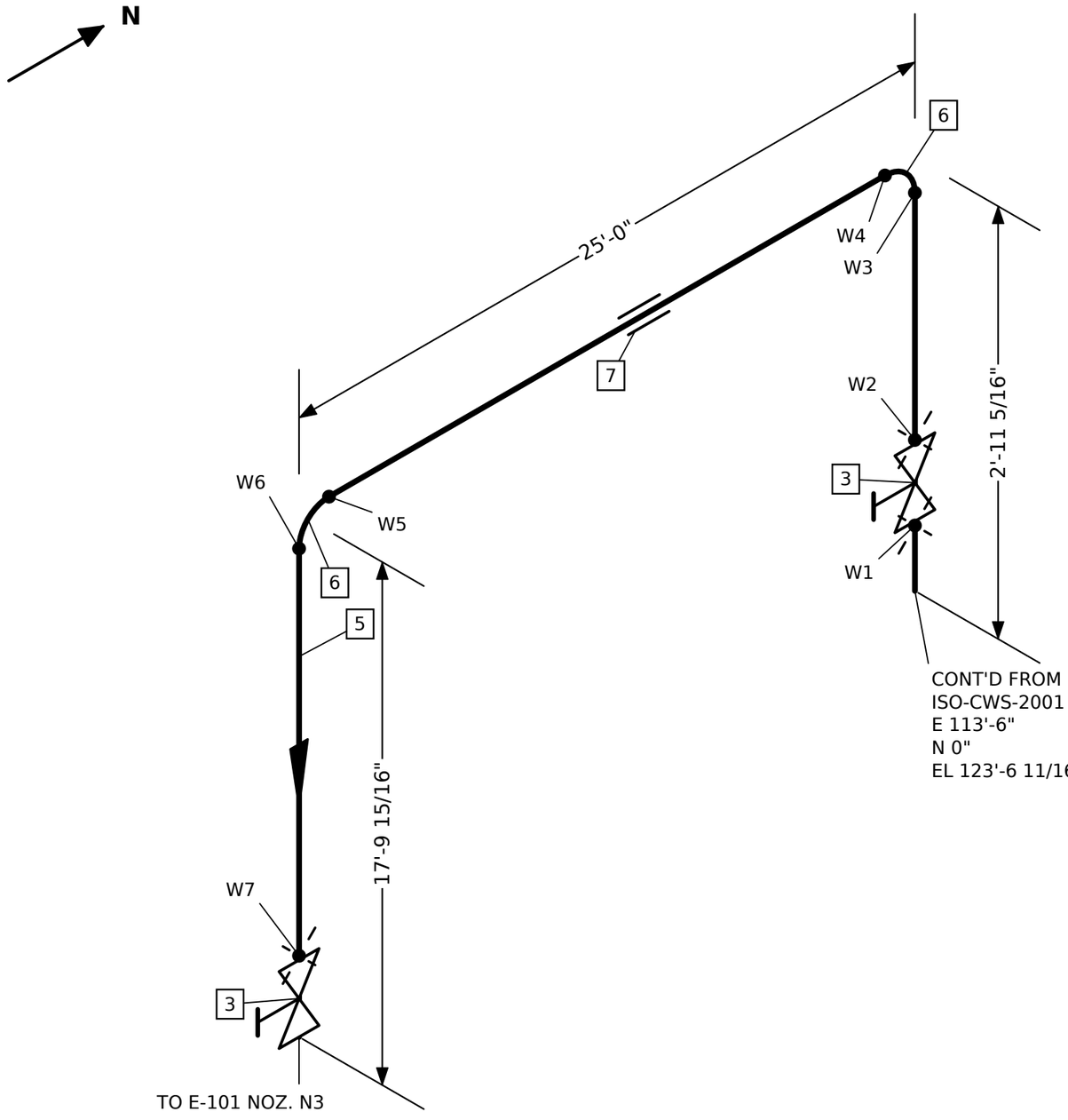
NOTES:
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	P

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
4"-P300-P-1005	H300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-P-1005	1 OF 4	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	3	4"	FLANGE WELD-NECK 4" P150 RAISED FACE
2	4	4"	GASKET 4" P150 COMPRESSED SHEET
3	2	4"	VALVE, GATE, WEDGE, OS&Y 4" P150
4	4	4"	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150
5	39'-7 5/16"	4"	PIPE 4" P150 BEVELLED ENDS
6	2	4"	ELBOW 90° R = 5/3 OD 4" P150
7	1	4"	HANGER FOR 4" LINE (HANGER-4x70.0)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

TEST PRESS. **-**	INSULATION **NONE**	PAINT **-**
PWHT **-**	NDE **-**	FLUID **CWS**

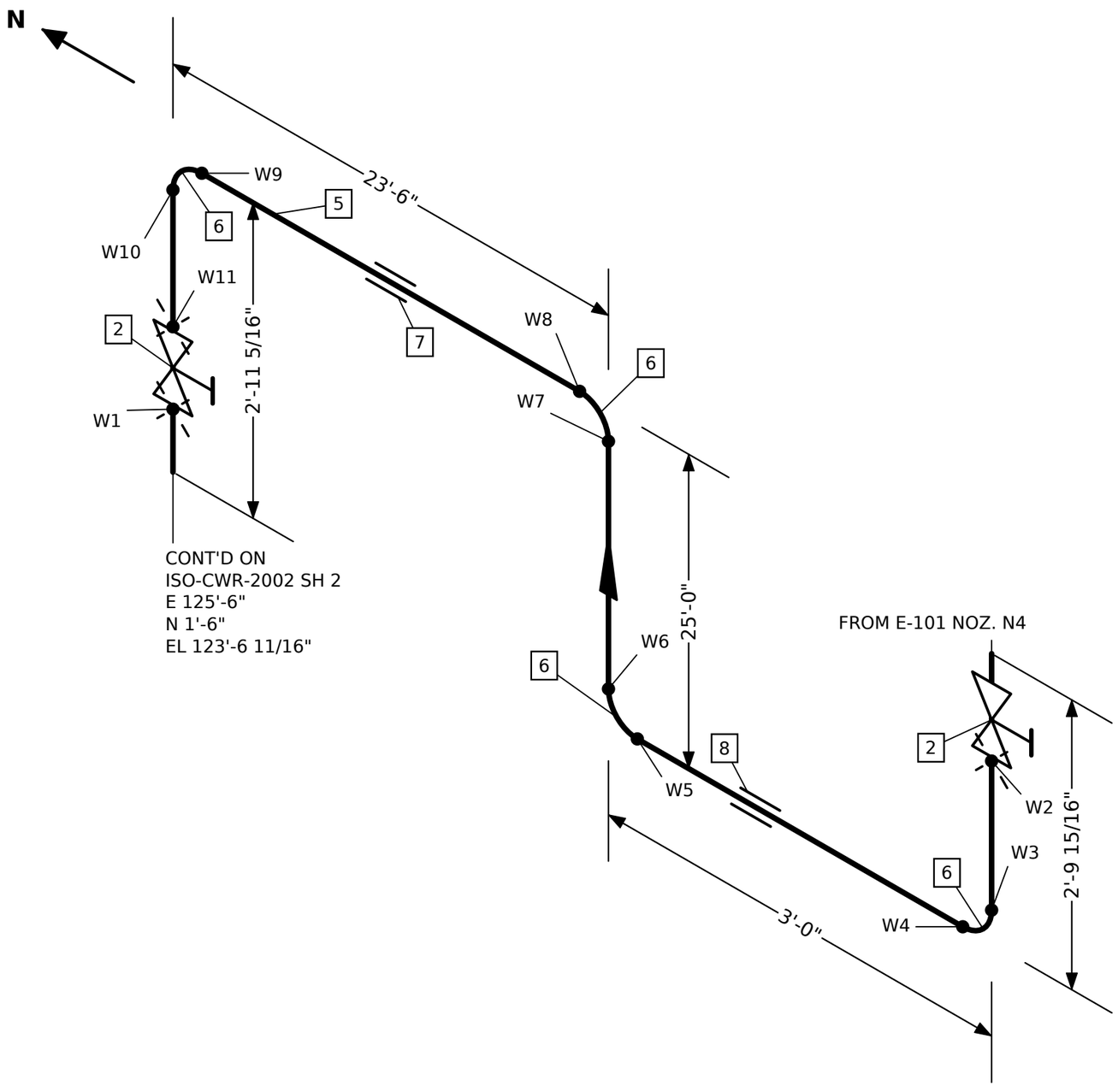
COMPANY

SMARTWELD

TITLE

WATER (COOLING AND PROCESS)

LINE NO. **4"-P150-CWS-3001**	PIPING CLASS **W150**	DESIGN PRESS. **150 PSIG**	DESIGN TEMP. **-**
DRAWING NO. **ISO-CWS-3001**		SHEET **1 OF 1**	REV **0**
DATE **2026-11-01**	DRAWN **-**	CHECKED **-**	SCALE **NTS**



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	4	4"	GASKET 4" P150 COMPRESSED SHEET
2	2	4"	VALVE, GATE, WEDGE, OS&Y 4" P150
3	4	4"	STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150
4	3	4"	FLANGE WELD-NECK 4" P150 RAISED FACE
5	48'-10 5/8"	4"	PIPE 4" P150 BEVELLED ENDS
6	4	4"	ELBOW 90° R = 5/3 OD 4" P150
7	1	4"	HANGER FOR 4" LINE (HANGER-4x48.5)
8	1	4"	TRUNNION FOR 4" LINE (TRUNNION-4x16.0)

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. REMOVE BURRS FROM ALL CUT ENDS
4. CHECK ORIENTATION BEFORE WELDING

TEST PRESS. -	INSULATION **NONE**	PAINT -	
PWHT -	NDE -	FLUID **CWR**	

COMPANY

SMARTWELD

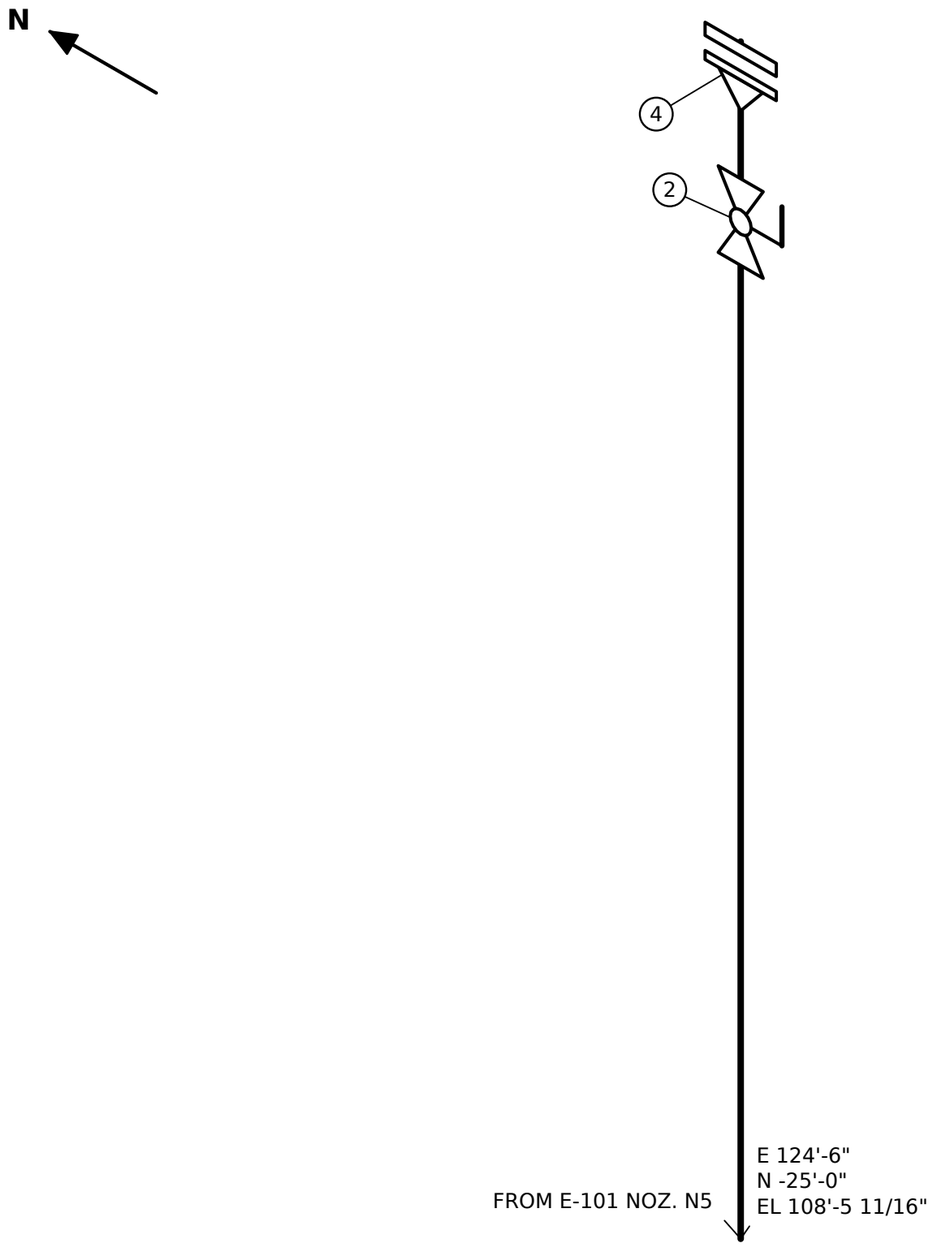
TITLE

WATER (COOLING AND PROCESS)

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
4"-P150-CWR-3002	**W150**	**150 PSIG**	**-**

DRAWING NO.	SHEET	REV
ISO-CWR-3002	**1 OF 1**	**0**

DATE	DRAWN	CHECKED	SCALE
2026-11-01	**-**	**-**	**NTS**



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	2	1"	GASKET 1" P150 COMPRESSED SHEET
2	1	1"	VALVE, BALL, FLOATING 1" P150
3	2	1"	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150
4	1	1"	FLANGE BLIND 1" P150 RAISED FACE

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. REMOVE BURRS FROM ALL CUT ENDS
4. PAINT PER PROJECT SPECIFICATION

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

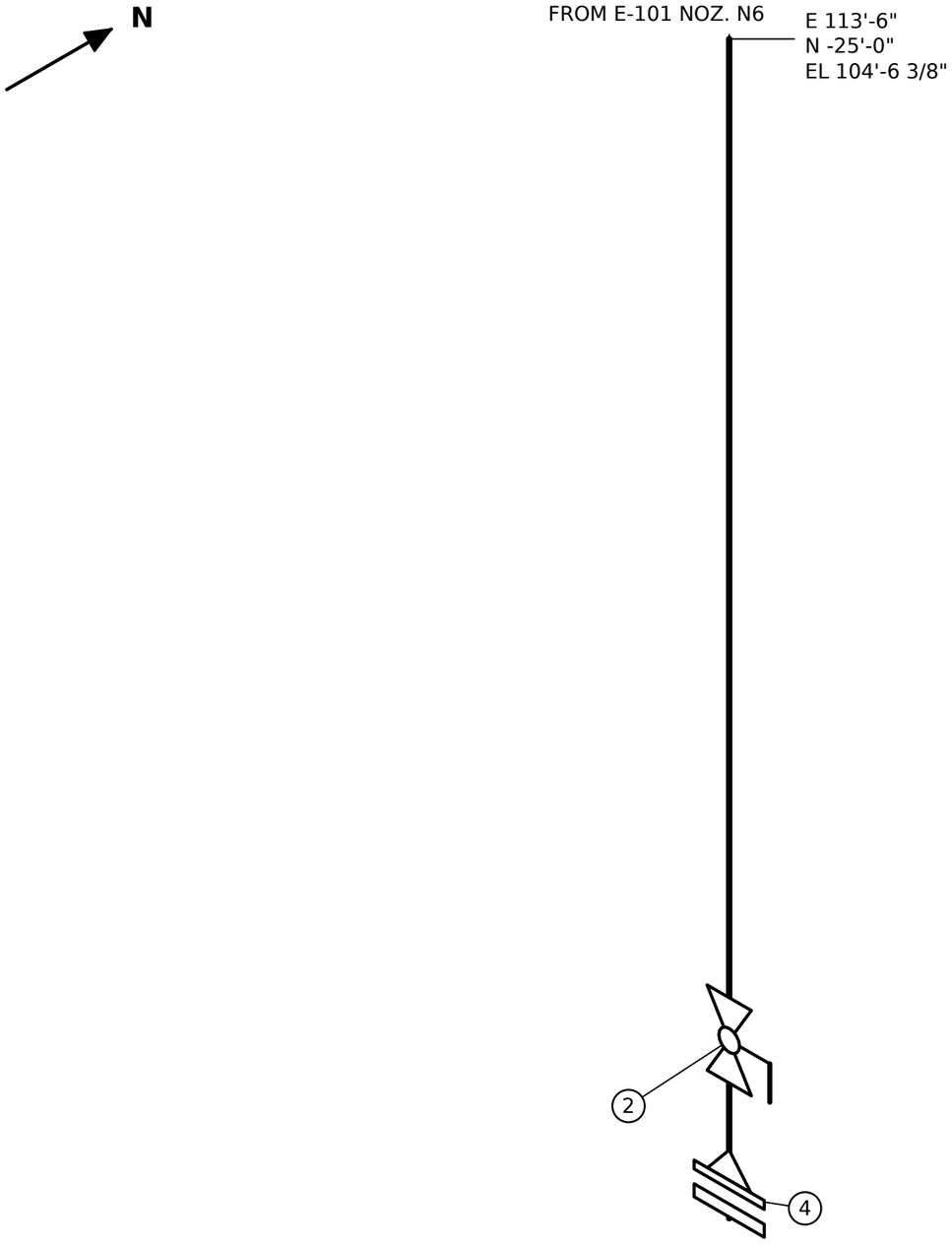
HYDROCARBON PROCESS

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	V

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
1"-P150-V-6021	H150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-V-6021	1 OF 1	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	2	1"	GASKET 1" P150 COMPRESSED SHEET
2	1	1"	VALVE, BALL, FLOATING 1" P150
3	2	1"	STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150
4	1	1"	FLANGE BLIND 1" P150 RAISED FACE

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

NOTES:

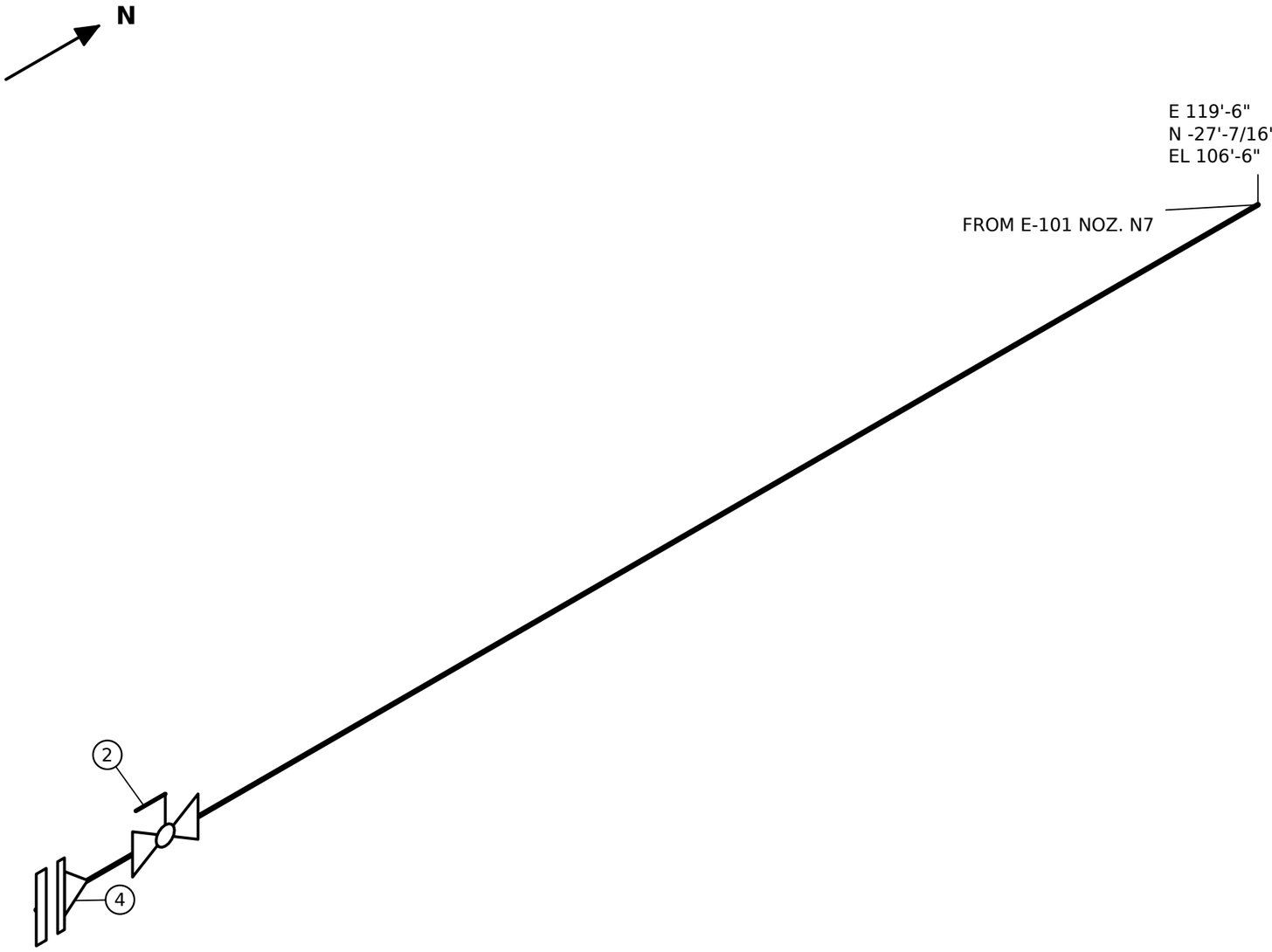
1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	D

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
1"-P150-D-6022	H150	150 PSIG	-

DRAWING NO.	SHEET	REV
ISO-D-6022	1 OF 1	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS



BILL OF MATERIALS

PT	QTY	NPS	DESCRIPTION
1	2	3/4"	GASKET 3/4" P300 COMPRESSED SHEET
2	1	3/4"	VALVE, BALL, FLOATING 3/4" P300
3	2	3/4"	STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300
4	1	3/4"	FLANGE BLIND 3/4" P300 RAISED FACE

NOTES:

1. ALL DIMENSIONS IN FEET AND INCHES UNLESS NOTED
2. BEVEL ENDS FOR BUTT WELD
3. CHECK ORIENTATION BEFORE WELDING
4. REMOVE BURRS FROM ALL CUT ENDS

REV	DATE	DESCRIPTION	BY
0	2026-11-01	ISSUED FOR CONSTRUCTION	-

COMPANY

SMARTWELD

TITLE

HYDROCARBON PROCESS

TEST PRESS.	INSULATION	PAINT
-	NONE	-
PWHT	NDE	FLUID
-	-	I

LINE NO.	PIPING CLASS	DESIGN PRESS.	DESIGN TEMP.
3/4"-P300-I-6023	H300	300 PSIG	-

DRAWING NO.	SHEET	REV
ISO-I-6023	1 OF 1	0

DATE	DRAWN	CHECKED	SCALE
2026-11-01	-	-	NTS

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